

Item 07 – Quantitative LLM Evaluation (Precision, Recall, Faithfulness)

Pareto Learning Guide (2026)

A Hiring-Oriented Curriculum by Expert Panel

1. Executive Summary

You can't improve what you don't measure. LLM evaluation is systematic testing at scale. This module teaches metrics, test set construction, and automated evaluation pipelines.



ML Research Scientist: *Evaluation is where research meets engineering. Good evaluation catches regressions before users do. Bad evaluation gives false confidence.*

- • Classification metrics: precision, recall, F1
- • Generation metrics: BLEU, ROUGE, faithfulness
- • Test set construction: representative, adversarial, edge cases
- • Automated evaluation pipelines
- • Portfolio: Evaluation framework with 100+ test cases

2. The Critical 20%

Concept	Why Critical	Deferred
Precision & Recall	Core classification metrics	Advanced statistical tests
Test Set Design	Quality of tests determines quality of system	Active learning for test selection
Faithfulness Metrics	Detect hallucination	Advanced NLI models
Automated Evaluation	Scale to 1000s of tests	Distributed testing infrastructure
Regression Detection	Prevent quality degradation	Statistical process control

3. Engineering-First Deep Understanding

3.1 Classification Metrics

For classification tasks (sentiment, intent, category):

- Precision = $TP / (TP + FP)$ - Of all positives predicted, how many are correct?
- Recall = $TP / (TP + FN)$ - Of all actual positives, how many did we catch?
- F1 = $2 * (Precision * Recall) / (Precision + Recall)$ - Harmonic mean

```
from sklearn.metrics import precision_score, recall_score, f1_score

predictions = ["positive", "negative", "positive", ...]
ground_truth = ["positive", "positive", "negative", ...]

precision = precision_score(ground_truth, predictions,
                           average='weighted')
recall = recall_score(ground_truth, predictions, average='weighted')
f1 = f1_score(ground_truth, predictions, average='weighted')
```

3.2 Faithfulness for Generation

Faithfulness: Does the generated answer only use information from the source?

```
def faithfulness_score(answer, source_chunks):
    # Extract claims from answer
    claims = extract_claims(answer)

    # Check each claim against sources
    verified = 0
    for claim in claims:
        if any(claim_in_chunk(claim, chunk) for chunk in
            source_chunks):
            verified += 1

    return verified / len(claims) if claims else 1.0
```

💡 **Applied AI Engineer:** *Faithfulness is critical for RAG. A 90% accurate but 50% faithful system is dangerous - it invents facts confidently.*

4. Day-by-Day Skill Reconstruction Plan

14-day intensive on evaluation.

- Day 1: Classification Metrics Deep Dive
- Day 2: Test Set Construction
- Day 3: Generation Metrics (BLEU, ROUGE)
- Day 4: Faithfulness Evaluation
- Day 5-7: Project - Evaluation suite for classifier
- Day 8: Automated Evaluation Pipelines
- Day 9: Regression Detection
- Day 10: A/B Testing Framework
- Day 11-14: Capstone - Complete evaluation framework

5. Common Blind Spots

- Only testing happy path (no edge cases, adversarial)
- Small test sets (<100 examples)
- No faithfulness checks (hallucination undetected)
- Manual evaluation only (doesn't scale)

6. Self-Assessment Questions

1. Q1: Your classifier has 95% precision, 60% recall. What does this mean?
2. Q2: Design a test set for a medical Q&A system.
3. Q3: How do you measure faithfulness automatically?
4. Q4: Your system regressed from 90% to 85% accuracy. Debugging process?

Expert Panel Synthesis

Dimension	Score	Evidence
Hiring Relevance	0.80	Evaluation crucial for production quality
Practical Applicability	0.85	Automated testing prevents regressions